

Trace properties at a glance

Seven identities, each linked to the section that proves it. What holds them together is that the trace sees only the diagonal — every entry below either exploits that or is limited by it.

UNCONDITIONAL

3

i **Linearity** $\text{tr}(cA + dB) = c \text{tr}(A) + d \text{tr}(B)$

UNCONDITIONAL

REQUIRES A, B both $n \times n$

The trace is a linear functional on the space of $n \times n$ matrices — it maps matrices to scalars while respecting addition and scaling. That is a stronger statement than the formula alone: it means the trace lives in the dual space, which is what makes the **Frobenius inner product** possible.

ii **Transpose invariance** $\text{tr}(A^T) = \text{tr}(A)$

UNCONDITIONAL

REQUIRES A square

Transposition reflects entries across the main diagonal, and the diagonal entries are exactly the ones fixed by that reflection. Since the trace reads only those, it cannot notice.

vi **Commutator trace** $\text{tr}(AB - BA) = 0$

UNCONDITIONAL

REQUIRES A, B both $n \times n$

Immediate from the cyclic property and linearity. The consequence is sharper than it looks: since $\text{tr}(I) = n \neq 0$, the identity matrix is **never** a commutator. No pair of matrices satisfies $AB - BA = I$ in finite dimensions — which is exactly why the canonical commutation relation of quantum mechanics needs infinite-dimensional operators.

CONDITIONAL

4

iii **Cyclic property** $\text{tr}(AB) = \text{tr}(BA)$

CONDITIONAL

HOLDS WHEN both products defined — A and B need not be square

COMMON ERROR reading it as full commutativity: $\text{tr}(ABC) = \text{tr}(BCA)$ but **not** $\text{tr}(BAC)$

Rotating the factors is allowed; permuting them arbitrarily is not. For three matrices the cyclic rotations $ABC \rightarrow BCA \rightarrow CAB$ all share a trace, while BAC generally does not. This is the identity everything below is derived from — it is the reason the trace survives a change of basis at all.

WHERE IT BREAKS

$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$
 $\text{tr}(ABC) = 1$ but $\text{tr}(BAC) = 0$

iv **Similarity invariance** $\text{tr}(P^{-1}AP) = \text{tr}(A)$

CONDITIONAL

HOLDS WHEN P invertible

A direct consequence of the cyclic property: rotate P^{-1} to the back and it meets P . The consequence is what matters — the trace is a property of the underlying **linear transformation**, not of the basis chosen to write it down. Two matrices with different traces cannot represent the same map.

v **Eigenvalue sum** $\text{tr}(A) = \lambda_1 + \lambda_2 + \dots + \lambda_n$

CONDITIONAL

HOLDS WHEN **eigenvalues** counted with algebraic multiplicity, over $\mathbb{C}$

Spectral information read straight off the diagonal, with no characteristic polynomial to solve. Over $\mathbb{R}$ the statement can fail because some eigenvalues may not exist there — a rotation matrix has trace $2 \cos \theta$ and no real eigenvalues at all. The sum is still correct once the complex pair is counted.

vii **Frobenius inner product** $\langle A, B \rangle_F = \text{tr}(A^T B)$

CONDITIONAL

HOLDS WHEN real entries; use A^H for the complex case

Gives the space of matrices a geometry — angles, lengths and projections, exactly as the **dot product** does for vectors. Positive definiteness holds because $\langle A, A \rangle_F$ is the sum of every entry squared, so the induced norm is the Frobenius norm.

Two of these are load-bearing for the rest. Linearity makes the trace a functional rather than merely a formula, and the cyclic property is what similarity invariance, the eigenvalue sum and the commutator identity are all derived from.